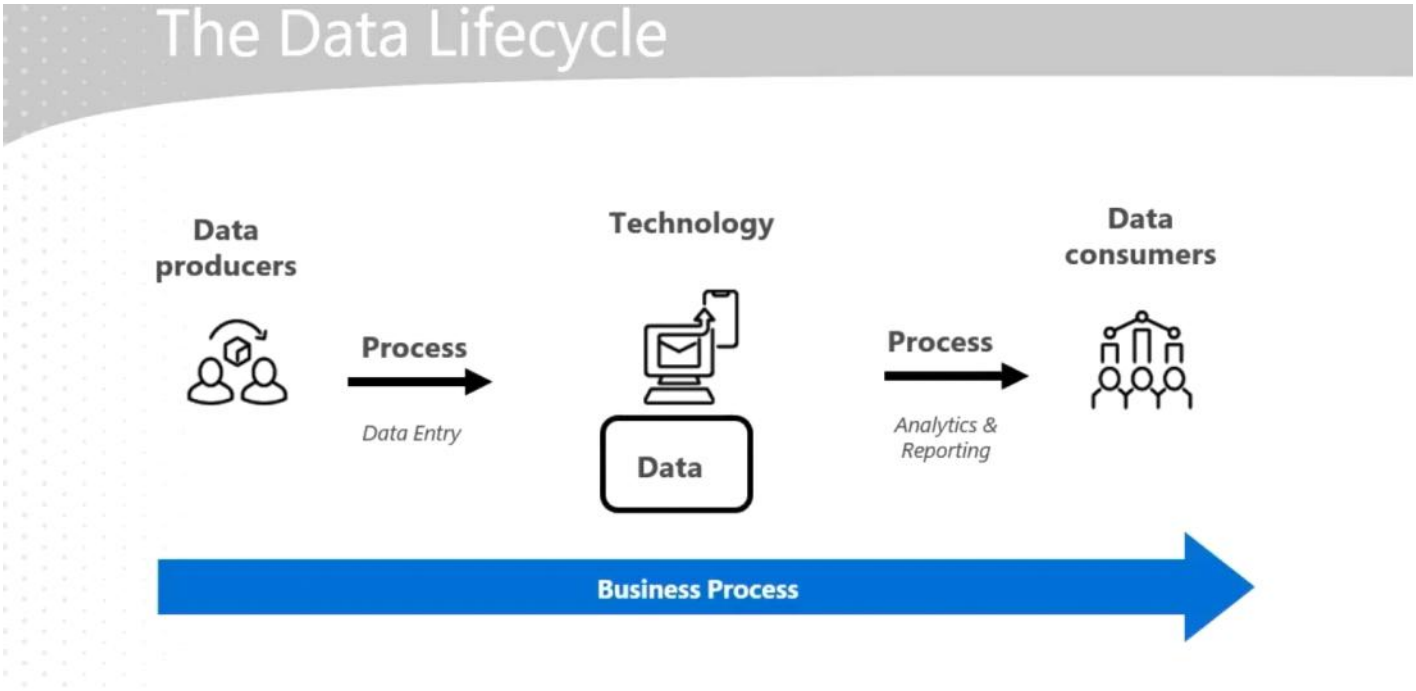
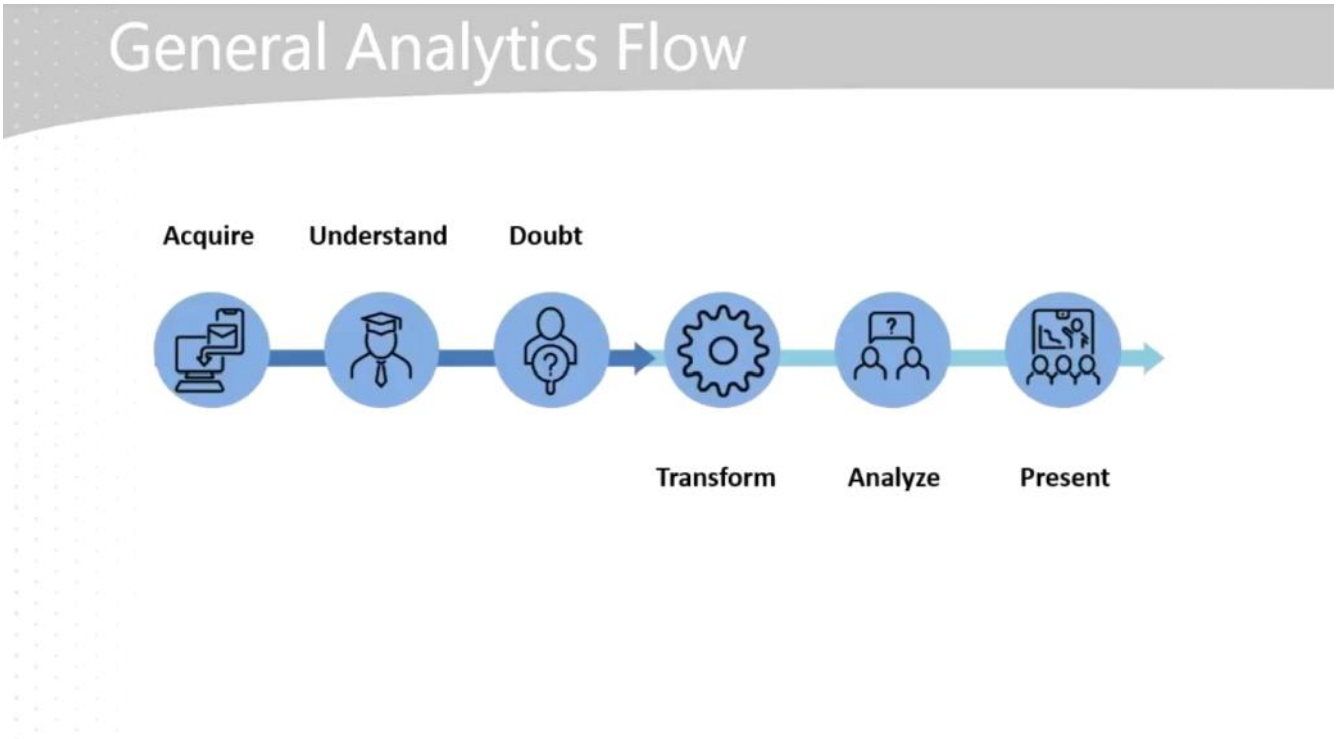
1.1 Understand the data lifecycle

Monday, January 26, 2026 11:13

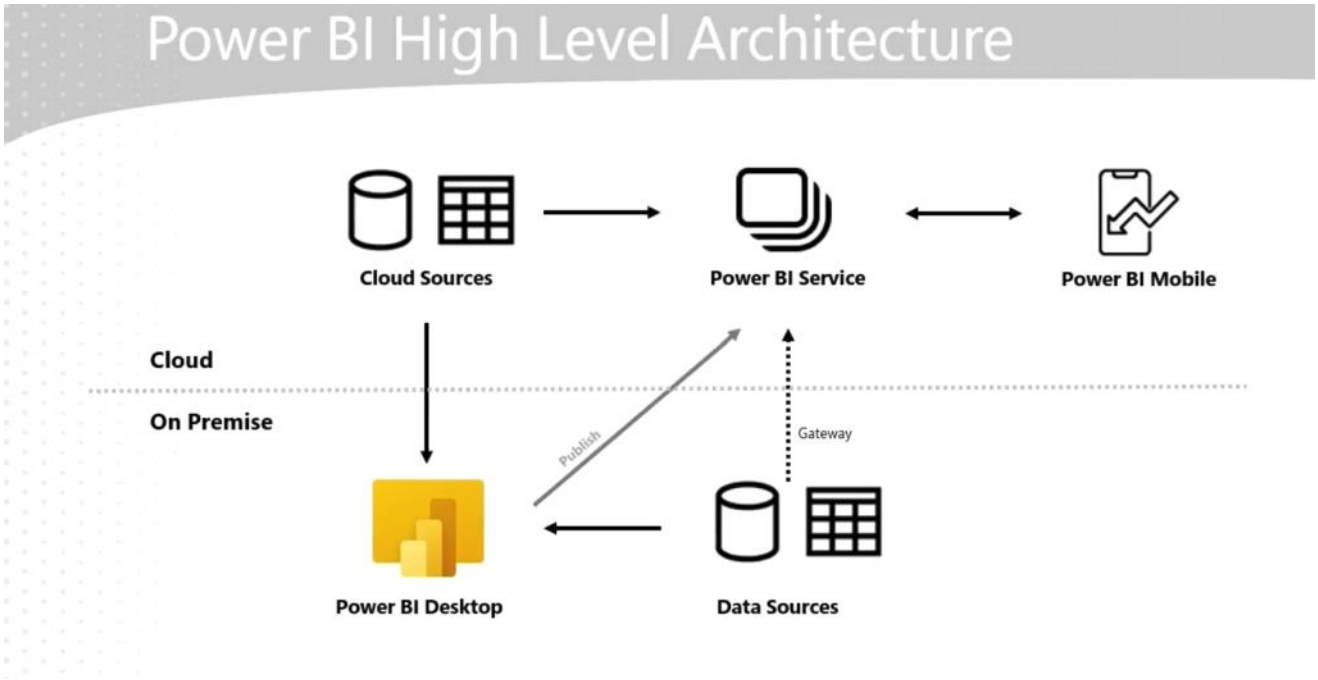


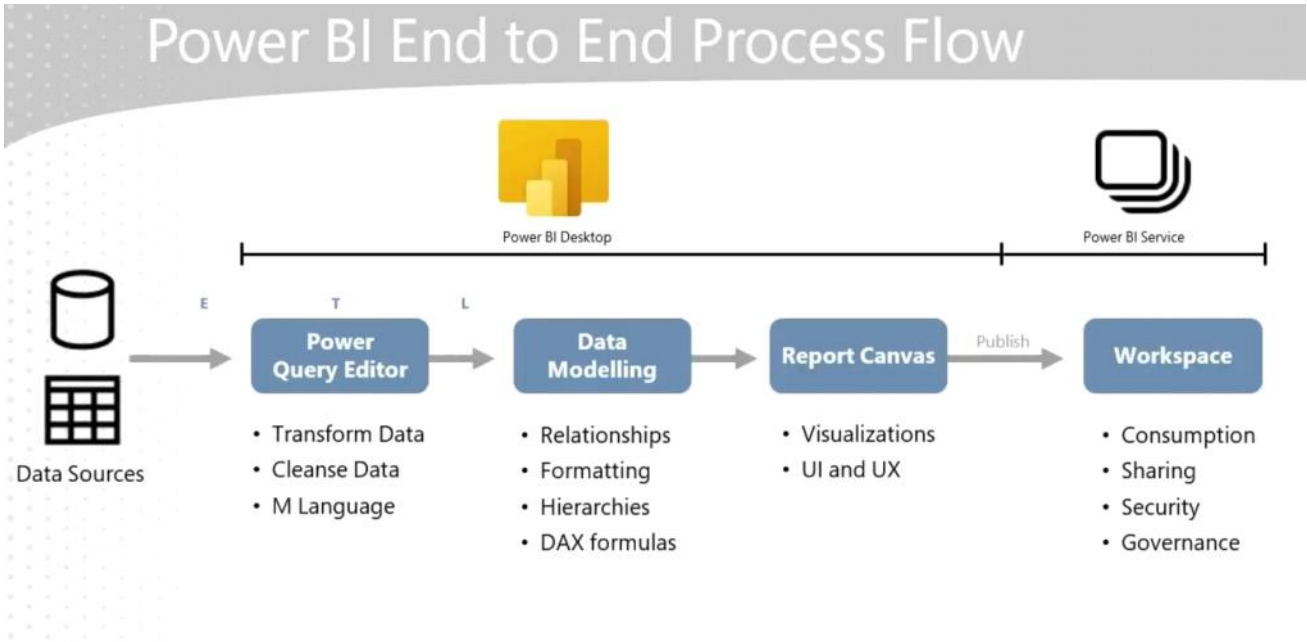
Data Producers can also be Data Consumers



1.2 Understand the Power BI environment

Monday, January 26, 2026 11:15



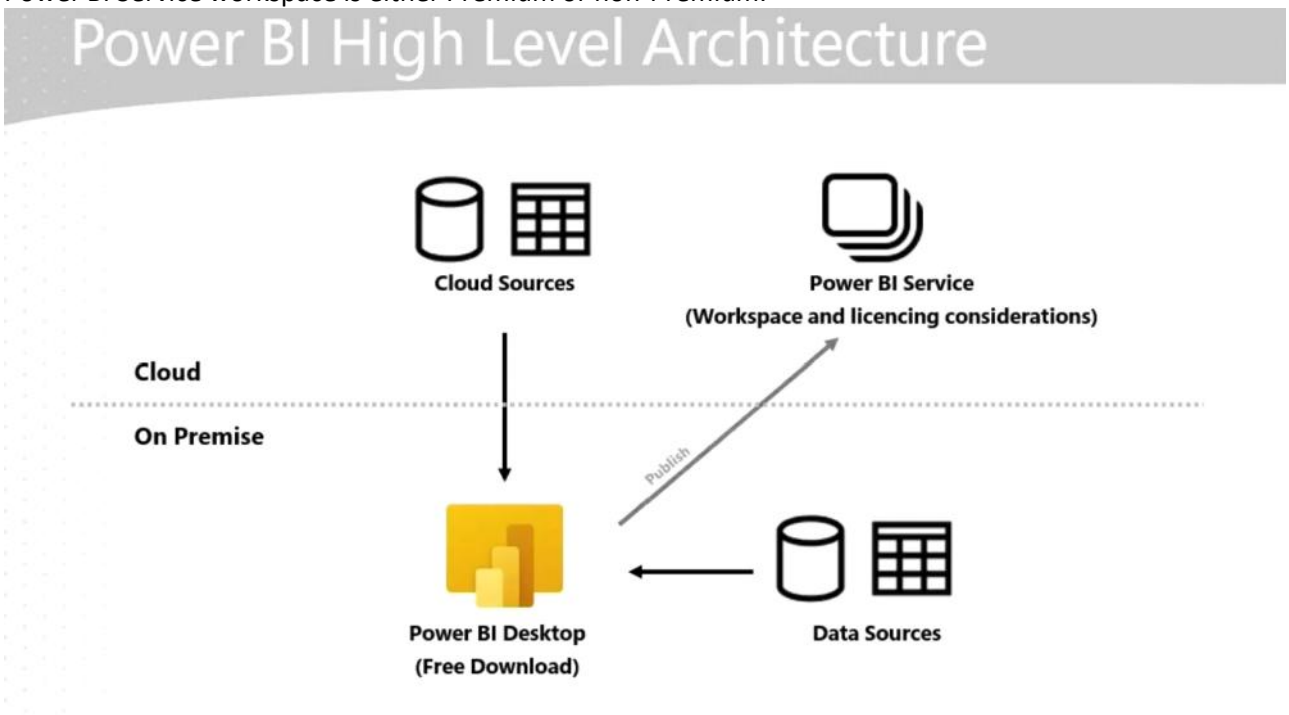


This is also how our course will flow.

1.3 Understand Power BI license

Monday, January 26, 2026 16:25

Power BI Desktop is free download.
Power BI Service offers individual either Free or Pro license.
Power BI Service workspace is either Premium or non-Premium.

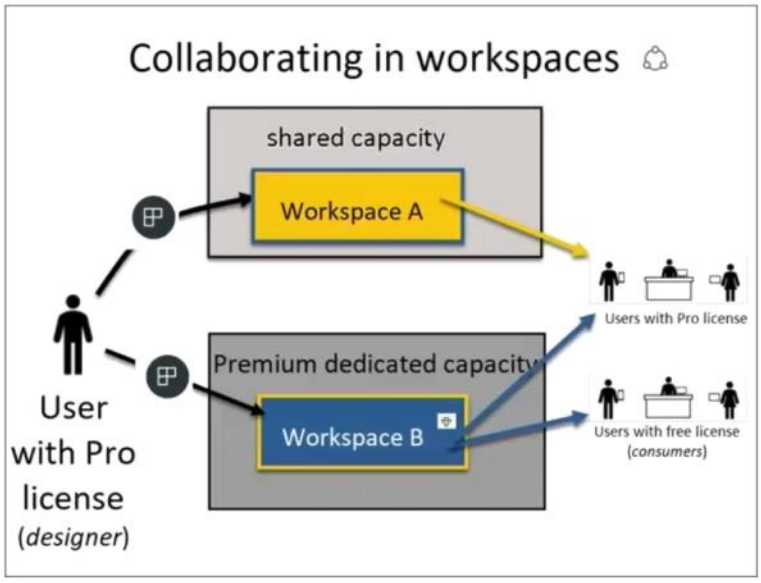


Licences and Capacity

	Not in Premium capacity	Premium capacity
Free	Use as a personal sandbox where you create content for yourself and interact with that content. A free license is a great way to try out the Power BI service. You can't consume content from anyone else or share your content with others	Interact with content assigned to Premium capacity and shared with you. Free, Premium per-user, and Pro users can collaborate without requiring the free users to have Pro accounts.
Pro	Collaborate with Premium per-user and Pro users by creating and sharing content.	Collaborate with free, Premium per user, and Pro users by creating and sharing content.

Source: https://learn.microsoft.com/en-us/power-bi/consumer/end-user-license

Scenario



Source: https://learn.microsoft.com/en-us/power-bi/consumer/end-user-license

1.4 Understand Power BI Release Cycles

Monday, January 26, 2026 16:32

Monthly Update Cadence

Power BI Desktop and Service

- https://learn.microsoft.com/en-us/power-bi/fundamentals/desktop-latest-update?tabs=powerbi-desktop

Power BI Mobile app

- https://learn.microsoft.com/en-us/power-bi/consumer/mobile/mobile-whats-new-in-the-mobile-apps

2.1 Power Query Editor Overview

Thursday, January 29, 2026 22:55

Query Best Practises

- >bring only the data that you need
- >remove rows that are not needed
- >remove columns that are not needed
- >rename columns using meaningful names
- >choose appropriate data types for columns

Rename Columns Best Pratices

- Give columns a proper name as soon as possible
- Three reasons we rename columns:
    - Better readability in self-service
    - Better results in Q&A
    - Consistency in the data model

Column Data Type Best Practices

- Give columns proper data types as soon as possible
- We select column data types based on:
    - How the column will be used
    - Storage considerations

Go self study the data types in Power Query

2.2 Import USA sales data

Thursday, January 29, 2026 23:01

Import from text/csv file
Transform
Give the table a meaningful name
Remove blank rows
Use first row as headers
Update data type of every column
Rename columns if necessary
Filter Rows by date
Save the pbix file named 'Sales'

2.3 Import Customer data

Friday, January 30, 2026 09:42

Import Customerlist text file
Rename the table as 'Customer'
Change the data type of a column
Replace value 1 and 0 with yes and no
Add a conditional column
Move the created conditional column
Create column from example by typing several examples of email addresses

2.4 Import Sales Territory Data

Friday, January 30, 2026 13:27

Import excel workbook by clicking 'new source' in power query editor
Rename the file as sales territory
Split column
Rename column where necessary
Verify the data type of each column
Trim unwanted spaces by choosing all related column and format>trim

2.5 Import Product Data

Friday, January 30, 2026 14:01

Import Product dimension excel from new source in power query editor
Change table name in the property tab to Product
It's better to pull the table instead of the sheet because even if you make some changes outside the table area, there is no impact on the data imported to power bi
Remove column by choosing the columns he want to keep; this is preferred way of doing this is explicitly telling the model which columns you need; it also prevents unwanted column 'slip-into' the power bi if someone else changed the data source.
Standardize column first before filtering, using 'capitalize each word'.
Filter the product subcategory not equal to xx.

2.6 Import Product Rollup Data

Friday, January 30, 2026 15:09

New source > txt/csv > product rollups

Reduce rows remove top number of rows

Use first row as headers

Standardise the subcategory entries with capitalise each word

Extract > Text After Delimiter, removing the PC- prefix

2.7 Merge Product and Product Rollup Queries

Friday, January 30, 2026 15:46

In product query, go Combine > Merge Queries
Choose Join Type

2.8 Import the other countries' sales data

Friday, January 30, 2026 16:59

Import from Folder

Filter out rows before 2020

Update column name and data type

2.9 Append Sales Queries

Monday, February 02, 2026 22:03

Home>combine>append queries>append as new

Romove the freight column

Update the column data type where necessary

Append queries explanation by animation

2.10 Create Columns

Monday, February 02, 2026 22:06

Obejective: how to create additional columns

Create total sales column in sales table

Choose the related columns and go to add column>standard>choose related formula>update column title

2.11 Model Loading and Cleanup

Monday, February 02, 2026 22:07

In power query editor
Right click on the table you do not want in the data model, unchoose 'enable load'; choose apply and close
In this lesson, the US_Sales, other coutries sales and product rollup was disconnected from the model.
Review the chart showing Power BI end to End process flow

2.12 Organise and document Power Query

Tuesday, February 03, 2026 22:30

Set up a group called facts in queries
Set up a group called dimensions
Set up a group called staging
Move united states sales to staging, move product rollup to staging, other coutnries sales to staging
Move sales to facts, customer, sales territory and product to dimension

Right click on product, go to description, type some description of the document in the description.

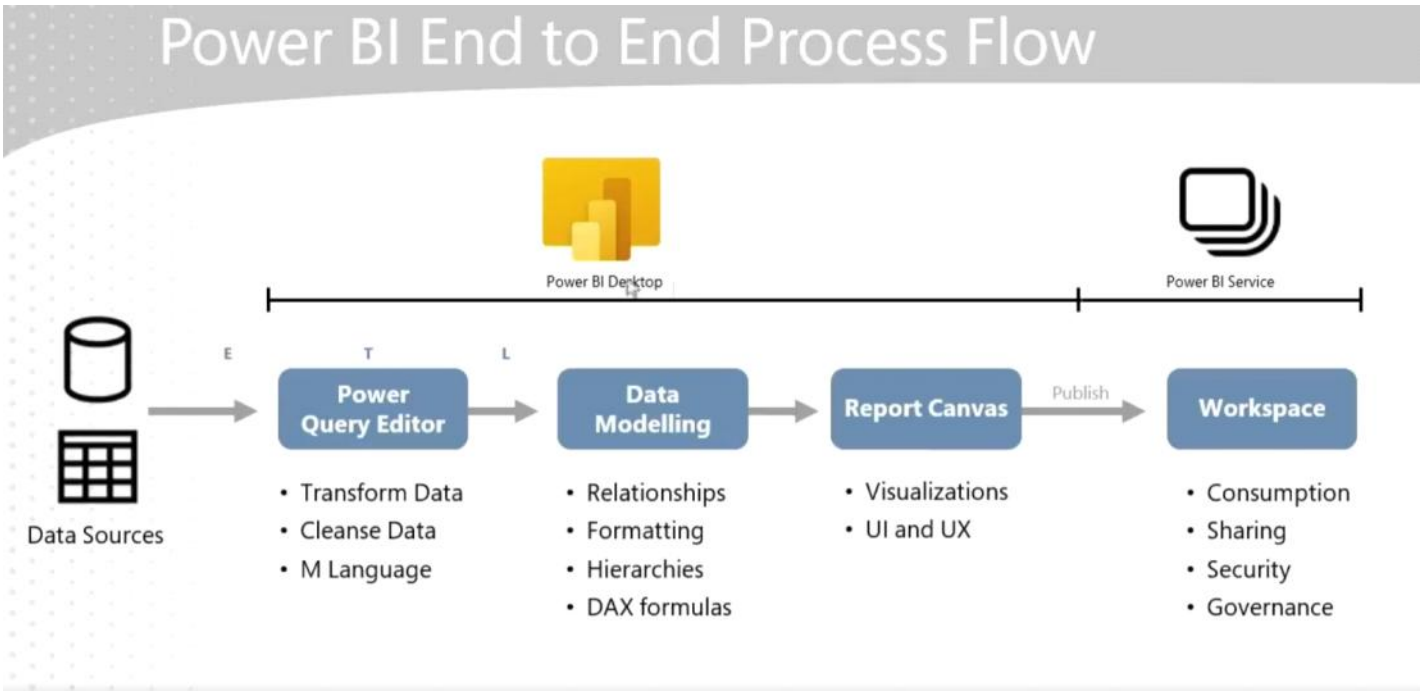
Query Dependencies
Try to understand how your model put together and where your data come from

Click on the data source settings in the home menu to see the data source

Go to advanced editor, to see the M code that was generated for your query.

3.1 Navigate the Power BI desktop

Tuesday, February 03, 2026 22:42



This lesson is about the second part of this chart.

The design and layout of Power BI is very similar to other Microsoft products.

First, the data pane on the right hand side. Then the different views.

3.2 Understand Dimensional Modeling

Tuesday, February 03, 2026 22:53

- Data Modeling Methodology that centres around measuring business processes.
- The Business Processes are known as Facts
- When we measure business processes, we need ways to describe those measurements.
- For this we use dimensions as report labels and/or filters.
- The combination of Facts and Dimensions is known as a **Star Schema** or **Dimensional Model**

Benefits of a Dimensional Model

- Make your data structures easier to navigate, manage and consume over time.
- Easier to layer in more information while keeping model results predictable
- Enable consistent analysis across an organisation
- Power BI is optimized to consume these models

3.3 Create Relationships and manage settings

Wednesday, February 04, 2026 22:02

Go to Model view
Rearrange the tables, put sales table in the middle, create star schema
Rearrange the position of the table in the model view does not affect the relationship among the tables
Then build relationships, click on manage relationships button, create 'new' relationship, choose the foreign keys or related columns, choose cardinality(many to one), cross filter direction single then close, one relationship is created

Power BI detects many to many relationship when building relationship between product table with sales, this is not ideal. To fix this, go back to the power query editor, look into the product table, there is duplicate rows where the product key is the same, other column also the same. Click on the reduce rows>remove duplicate rows, the product table is de-duplicated.

Go back to the model view, click manage relationship again. The cardinality went from many to many to many to one.

When creating relationship between sales and sales territories, he found the column of the two tables are different. Thus, going back to the power query again, update the column name with the ones in sales table, close and load.

Ideally, we want our filters to flow from the dimensions to the sales.

Definition of Cardinality:
>Cardinality's official, non-database dictionary function is mathematical: the number of values in a set.
>When applied to databases, the meaning is a bit different: it's the number of distinct values in a table column, relative to the number of rows in the table. Repeated values in the column don't count.

Relationship Direction:
Whatever direction of the relationship is, that determines how Power BI filters the data.

3.4 Modify Column Properties

Thursday, February 05, 2026 09:41

Go to Table View
Go to Sales Territory > Choose Sales territory country > Column Tools> Data Category> Country
In Data Plan, the sales territory country has a globe icon beside it.
Go to Customer Table > Column Tools > City, country…
Go to every column that can be categorized geographically

Go to sales table> Column Format> Currency> choose United States Dollar, do this to all currency related columns

Go to customer> choose yearlyincome > column tools > choose 'don't summarize', as this is a column you don't want to summarize
Go to Sums, check each number columns to see if summarization make sense for each column. If it's sum or average, etc.

3.5 Create a Column Group

Thursday, February 05, 2026 15:19

Build data groups:
Table view> Sales Territory> Column tools> data group
Put row value into groups and give name to the group
Include other group
New column group created

3.6 Create Hierarchy

Thursday, February 05, 2026 15:54

Sample Date Hierarchy

Sample Product Hierarchy

Go to model view> product table> open properties and data pane> product category > righ click and create hierachy > name it product Hierarchy> Category - Subcategory- product name

Hide the category, subcategory and product name columns, if we go to report view, I will only see these column under the product hierarchy

3.7 Optimize Relationships

Friday, February 6, 2026 10:51 PM

Go to table view
The connection between sales and sales region is not optimal as they are based on the text columns
Merge sales and sales territory and introduce sales territory key in the sales table
Remove the relationship between the two table
Set up a new relationship based on the sales territory key.

3.8 Discuss data model clean-up

Saturday, February 7, 2026 10:04 PM

There are columns in the models that we do not want the users to see.
Choose the customer key, product key and sales territory key and hide them.

4.1 Understand DAX

Saturday, February 7, 2026 11:05 PM

Data Analysis Expressions
Modeled after Excel formula language
Functional Language
Used to encapsulate business logic in your data model
Examples:

- Calculate a full name from a first and last name
- Calculate Gross Margin %
- Calculate Year over Year % Changes

Full Name = Customer[FirstName] & " " & Customer[LastName]

- First part of the formula is the name of the column or measure that is being created.
- The name can contain spaces.
- Followed by an equal sign.
- This returns the value to the right of the equal sign.
- In this example, we are creating a calculated column.
- It concatenates the first name and last name from the customer table.
- The reference to customer[FirstName] is known as a fully qualified name.
- Table names are required whenever the column is from a different table than the current table. Table names must be unique within the database.
- Table names must be enclosed in single quotation marks if they contain spaces, other special characters or any non-English alphanumeric characters.

Example:
Created a calculated column in customer table.

4.2 Create basic DAX expressions

Sunday, 8 February, 2026 22:16

About Calculated Columns

Calculated on a row-by-row basis.
Creates a new physical column.
Take up space in the data model.
Value is calculated at data load time.
Typically reside in dimension tables to be used as labels on reports and/or slicers.
Can often be pushed into the Power Query engine.

Eg: Line Total Sales = Sales[UnitPrice] * Sales[OrderQuantity]

About Measures

An aggregate of a column.
Do not take up space in the model.
Value is calculated at run time.
Typically reside in fact tables.
Belong to the data model.

Eg: Total Sales = SUM(Sales[Line Total Sales])

Row Context Defined

Row Context is applied as a virtual pointer which moves down a table of data row by row and is used in a calculation.

Filter Context Defined

This context is applied by taking all filters that surround a DAX expression and then applying the filters to the expression to affect its result.

More on Context

Some areas where context can be applied or affected:

- Within a visual
- Visual level
- Page level
- Report (all pages) level
- Inside an expression
- Visual interactions

What is in a pbix File?

Sales.pbix

Dataset	Report		
	Page 1	Page n	
	Visual 1	Visual 1	Visual 2

4.3 Use Logical expressions in DAX

Sunday, 8 February, 2026 22:33

Go straight into the Power BI demonstration by creating:

New column in customer:

```
Primary Buyer = AND(Customer[Car ownership] = 0, customer[home ownership] = "No")
Secondary Buyer = OR(Customer[Car ownership] = 0, customer[home ownership] = "No")
```

New Column in product:

```
Color Category = IF('Product'[Color] IN {"Black", "Blue", "Red"}, "Darker", "Lighter")
```

4.4 Comprehensive create calculated column and measures: Part one

Sunday, 8 February, 2026 23:11

Created several more calculated column in sales table: line product cost, line margin, line margin %
Added these columns in the report view

4.5 Comprehensive create calculated columns and measures - 2

Wednesday, 11 February, 2026 21:56

In the sales table, add two measure on top of the measures previously created: Total Margin and Total Margin percentage
Adding these two measures to the table also eliminates the need for columns like line margin etc. uncheck these columns in report view and keep only total sales, total margin and total margin percentage.

4.6 Comprehensive create calculated columns and measures - 3

Wednesday, 11 February, 2026 22:06

Added another measure called Total Sales SUMX, using the SUMX function, this function uses a combination row context and the filter context, by calculating the expression in the function and adding them row by row.

4.7 Use the FILTER function

Wednesday, 11 February, 2026 22:35

Report View, create the following measures:
1.Count number of rows of sales line items, which is the rows of the sales table, use COUNTROW()
2.Count number of sales orders, use DISTINCTCOUNT()
3.Count number of sales line greater than 50, use COUNTROW(FILTER(Sales, Line Sales >50))

4.8 Use ALL Function

Thursday, 12 February, 2026 22:51

In report view, create the following two measures:
1.COUNTROWS(ALL(Sales))
2.Divide COUNTROWS(Sales) by the previous measure, showing each years order line quantity percentage.

5.1 Build a date Table

Friday, 13 February, 2026 11:20

In Table view, Create New Table:

```
Order Date = CALENDAR(MIN(Sales[Order Date]),MAX(Sales[Order Date]))
```

Then in the same Date table, create the following columns:

```
Order Date Year = FORMAT('Order Date'[Order Date], "YYYY")

Order Date Month = FORMAT('Order Date'[Order Date],"MMM")

Order Date Month Number = MONTH('Order Date'[Order Date])
```

Set data type to Date, Whole number, Text and Whole number

And remember to 'Don't summarize' all these columns.

5.2 Disable the auto date/time setting

Friday, 13 February, 2026 11:32

The lecturer wants to do this because in every data type 'Date', Power BI will automatically create a 'Date Hierarchy' under the column, which will make the model heavier.

He suggest the user should turn off this function.

Go to Power BI option & setting, choose option, then Current file > Data Load, uncheck 'Auto Date/Time'.

5.3 Designate a date table

Friday, 13 February, 2026 12:34

In the table view, choose Order Date table, in table tools, enable mark as date table.

5.4 Build Relationships between order date and sales

Friday, 13 February, 2026 14:07

Model View, connect the order date and the sales table, one to many relationship.

5.5 Basic Time Intelligence Functions

Friday, 13 February, 2026 14:52

Create a duplicate page of the report.

Remove the order date column and add in the date table year and month column.
In table view, go to date table, sort the column by month number, go back to report view, the month will be sorted by chronological order.

Remove all other columns except for the Total Sales. Add the following measures and format it in currencies:

```
YTD Sales = TOTALYTD([Total Sales],'Order Date'[Order Date])

QTD Sales = TOTALQTD([Total Sales], 'Order Date'[Order Date])
```

And add them to the report.

5.6 Use the CALCULATE Function

Friday, 13 February, 2026 16:19

'Turbo Charged sumif() function'

Add in the report view the following measure:

```
PY Total Sales = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Order Date'[Order Date]))
```

6.1 Use Quick Measures

Friday, 13 February, 2026 22:30

www.daxformatter.com

Use quick format > time intelligence > year over year change
Drag Total sales and order date to the pane, the following DAX is created

```
Total Sales YoY% 2 =
VAR __PREV_YEAR = CALCULATE([Total Sales], DATEADD('Order Date'[Order Date], -1, YEAR))
RETURN
    DIVIDE([Total Sales] - __PREV_YEAR, __PREV_YEAR)
```

7.1 Navigate the Power BI Desktop Report View

Friday, 13 February, 2026 22:44

Visualisation Pane
Build Visual/Format
Format> Page Information/Canvas Settings/Canvas Background/Wallpaper/Filter pane/Filter Cards

Filters Pane

7.2 Create a clustered column chart

Friday, 13 February, 2026 22:48

Build a dashboard.

Line chart, total sales by continent bar chart, margin % by product category bar chart, and two slicers.

This section creates the visual of total sales by continent, and went through the visual setup of the bar chart.

7.3 Create a stacked bar chart

Tuesday, 17 February, 2026 22:35

Actually created a clustered bar chart
X Axis, Total Margin %,
Y Axis, Product Subcategory

Adjust 'Visual' and 'General' Pane of the chart.

7.4 Create a line chart

Tuesday, 17 February, 2026 23:02

X-Axis: Orderdate Year
Y-Axis: Total Sales

Adjust 'Visual' and 'General' Pane of the chart.

7.5 Create a KPI Visual

Thursday, 19 February, 2026 9:23

Create a new measure in sales table: Target = 7000000
In real scenarios, the target will be based on regions/products, here we do it for simplicity.

Choose 'KPI' visual in the visual table, total sales in the Value pane, order date year in the trend axis, target measure in the target pane.

Format pane, fine tune the visual parameters like font size, switch off distance to goal, etc.

7.6 Create a card Visual

Thursday, 19 February, 2026 9:32

Choose 'Card' visual in the visualization pane, Choose Target measure in the fields

Format pane, change font size, turn off category label, General sub tab change size and position, put 'Target' in title.

7.7 Create Slicers

Thursday, 19 February, 2026 9:37

Click 'Slicer' in the Visuals, put sales territory continent in the field, go to format, slicer setting, choose drop down in the optinos, select multi-select in the selection and choose 'select-all' option, change font size to 14, and bold, General sub tab, size and position tune.

Copy the first slicer vicual, and paste, and then work on the duplicate. Change the field from continent to order date year, rename the slicer from continent to year; General sub tab, size and position tune.

Open Filters pane, you can use this pane to do the slicers, however it depends on the preference also with the users of the visual.

8.1 Add a Canvas Background

Thursday, 19 February, 2026 14:30

Format > Canvas Background, upload the background.png

Page 1, Canvas background, upload dashboard background 3.png. Change page name to DAX Demos.

Page 2, Canvas Background, dashboard background 2. Change page name to product sales details.

8.2 Apply a theme

Friday, 20 February, 2026 10:29

Overview page, go to view, choose theme, import theme from source file

This is widely used in the corporate world to incorporate CI or VI.

8.3 Adjust existing Visuals

Friday, 20 February, 2026 10:43

Overview Page, choose the Total Sales visual, turn off target and distance to goal.

DAX Demos Page, hide 'count of all sales order line item', also remove this column from visual, also remove total sales SUMX, rearrange the columns to make it fit the canvas

Product Sales Details Page, space the column more evenly.

8.4 Create a custom Tooltip

Friday, 20 February, 2026 10:54

Create a new page, call it 'sales regions'

Go to format, choose 'allow use as tooltip', go to canvas setting, make sure it's type is tooltip
Go to view ribbon, page view, choose 'actual size'

Choose pie chart, fit the pie full in the canvas, drag 'country' in the legend, and 'total sales' in the value.

Go to format of the pie chart, switch off legend, go to detail labels, label contents choose all detail labels, go to value subtab, display units in millions and decimals as 1, general tab, title rename to total sales by country, bold, under the effects, ensure the visual border is off.

Go to overview page, choose the total sales by continent visual, go to format, general, use as tooltip, choose sales region in the page dropdown box, and report page in the type dropdown box.

8.5 Explore drill through

Friday, 20 February, 2026 11:20

Go to product detail page, delete QTD sales, choose product subcategory under product category, remove product category, rearrange column order by bringing up product subcategory, drag product category to the drill through section under the build visual pane.

Go to overview page, choose margin % by product category, right click and choose drill through

Hide the product sales detail and sales region page.

8.6 Apply Conditional formatting to a Visual

Friday, 20 February, 2026 22:21

Choose product sales details visual, go to Visual > Cell Elements, choose series name Total Sales YoY%(this is the column we are going to put conditional formatting on). Switch on 'Icons', choose 'fx', apply corresponding settings and the selected column starts to show traffic lights according to the setting.

Update some cosmetics of the same visual, go to format, visual, in style presets choose alternative rows, go to column headers and totals, change the background color to desired one.

8.7 Build an order hierarchy

Friday, 20 February, 2026 22:27

Go to model view, choose order date table, right click order date year, in its property pane, change name to order date hierarchy, in the same property pane, in the bottom find hierarchy, choose order date month and order date to add level. Hide the actual order date, order date month and order date month number, order date year columns. Only keep the order date hierarchy.

8.8 Enable a drill-down on a visual

Friday, 20 February, 2026 22:46

In overview page, choose sales performance visual, remove the order date year column in the x-axis, drag the order date hierarchy in. try out the different drill down and up menus on top of the visual.

8.9 Edit Interactions between Visuals

Friday, 20 February, 2026 22:50

Click on year slicer, go to format ribbon and click edit interactions. Choose either 'filter' or 'none' on top right of each visual. Do the same thing on continent, for example, choose 'none' for total sales by continent. For non-slicer visuals, you can choose 'filter', 'highlight' or 'none'.

8.10 Use the Selection Pane

Friday, 20 February, 2026 22:55

Overview page, View ribbon, Selection pane

Selection pane allow you to choose visibility and layering orders of subjects in a report.

Reorder the 7 subjects in the tab-order sub pane, tab-order literally means the order of choice of subjects if you tab the 'tab' key on the keyboard.

Hide the year and continent slicer in the layer order.

8.11 Use the bookmarks pane

Friday, 20 February, 2026 22:58

Overview page, make sure the visuals are completed unfiltered, click add in the bookmarks pane, rename the bookmark as sales home, so this is the default page.
Go to Continent slicer and choose north america, and year choose 2021, click on 2021 in the sales performance visual to drill down. Hide the continent and year slicers. Click add bookmark and rename to north america sales 2021.
Repeat the second step but filter the year to 2022, add bookmark and rename to north America sales 2022.
Choose the sales performance visual and drill down to month view of 2022, right click on the bookmark 2022 and choose update, the third bookmark is then updated.
Each bookmark is a page where you develop your stories, it's working like a PPT.

8.12 Explore the Filter Pane

Saturday, 21 February, 2026 22:45

We don't train much on this topic in this level 1 course.

We could use the filter pane to replace certain slicers.

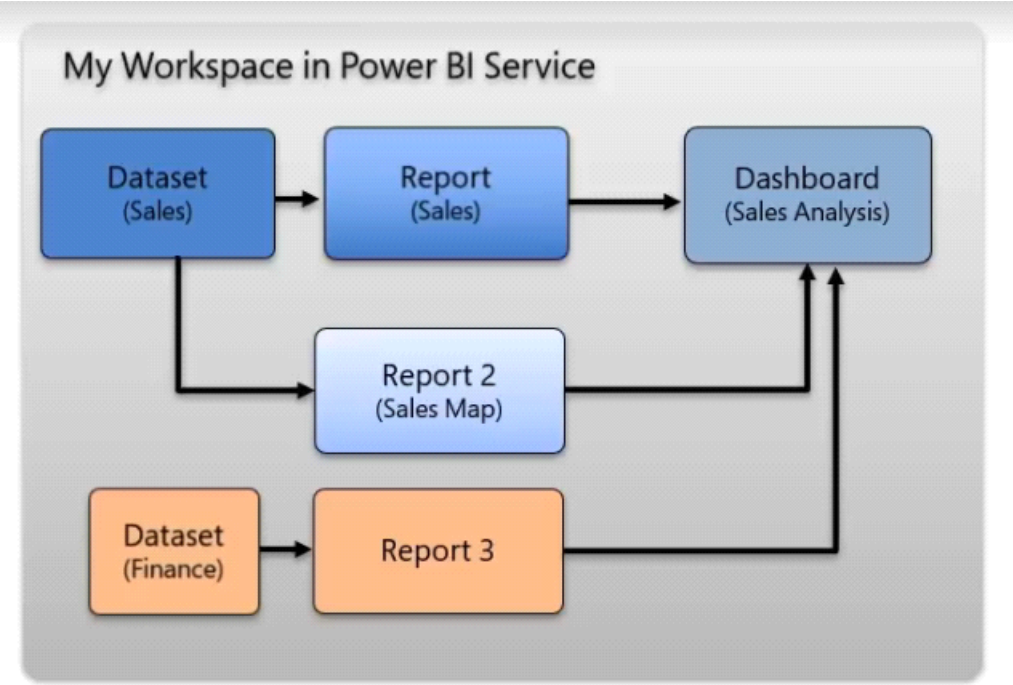
8.13 Explore the structure of a .pbix file

Friday, 20 February, 2026 17:27

The separation of dataset and report visuals are one of the key differences between Power BI and Excel.

10.1 Create a new report using an existing dataset

Sunday, 22 February, 2026 22:06



You can use power bi service like the desktop to create a report using the same dataset.

10.2 Build a dashboard

Sunday, 22 February, 2026 22:11

My workspace, new > dashboard
Name it sales analysis, pin the sales map to the sales analysis,
Go to sales report, overview page, pin the total sales card and the sales performance, total sales by continent to the sales analysis dashboard.

10.3 Configure the Dashboard

Sunday, 22 February, 2026 22:17

Sales Analysis Dashboard

Sales Map > Tile Details, Change name to National Sales Map

You can edit the desktop layout, as well as mobile layout.

10.4 Set an Alert

Sunday, 22 February, 2026 22:20

Sales Analysis, Total sales card, Manage Alerts, add alert rule.

The alert is only intended to the owners. You can use the Microsoft power automate to alert others.

10.5 Use the Q&A Dialog

Sunday, 22 February, 2026 22:25

'Ask a question about your data', on the top of your dashboard.

For example, type in 'total sales', power BI will come back with the number from the dataset.
Type 'total sales by product name' power BI will respond with a bar chart.
You can work with the visual the Q&A created like any visuals in report.

Naming conventions is important to get good response here.

11.1 Create row-level security (RLS) roles

Sunday, 22 February, 2026 22:31

Modeling tap, manage security roles,
Create new roles, north America Controller
Select Tables,
Filter data > Add Rule > Show data when…

Create new roles, Australia Controller
Filter data > Add rule > Show data when… Sales territory = Australia

11.2 Validate RLS Roles using Power BI desktop

Sunday, 22 February, 2026 22:38

Modelling Tab, View as > Choose Australia Controller
The power BI then shows the report from the view of the Australia Controller.

Go to Model View, we can see the sales territory dimension is forcing the model to show only the data of Australia.

Go to Report view, home tab, make sure the report is in original state and Publish the report again

11.3 Assign and Validate Roles in the Power BI Service

Monday, 23 February, 2026 22:25

Power BI service, my workspace, go to sales dataset 'more options', choose security

Click North America Controller, now viewing the report as 'North America Controller', the report is only limited at North America data

Go back to 'Row-Level Security', choose one of the controllers, assign person to the role. You can also click on 'test this role'.

12 Summary

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- Getting Started with Power BI.
- Importing and Transforming Data using Power Query Editor.
- Enhancing the data Model - basics
- Enhancing the data Model - DAX Basics
- Enhancing the data Model - DAX Time Intelligence
- Enhancing the data Model - Quick Measures
- Building Desktop Visualizations - The Basics
- Building Desktop Visualizations - Next Level
- Using the Power BI Service - Reading View
- Using the Power BI Service - Next Level
- Implement Security and Access